10f. SIMD: Branchless Code

Martin Alfaro
PhD in Economics

INTRODUCTION

SIMD accelerates workloads by applying a single instruction to many elements at once. Its effectiveness critically hinges on all elements following the same control flow. Yet, this pattern can be disrupted when conditionals are introduced, either explicitly in the source code or implicitly by the compiler. And, while the compiler attempts to mitigate the issue by transforming code into SIMD-friendly forms, these adaptations tend to introduce additional work that reduce the net benefit of SIMD.

The current section explores strategies for efficiently applying SIMD in the presence of conditional behavior. In particular, we'll show how to restructure computations to make control-flow branches disappear, allowing the program to preserve a uniform instruction stream.

We'll first examine scenarios in which the compiler introduces branches as a byproduct of its internal computation techniques. In these situations, users may be unaware that conditional control flow has been generated, making it essential to recognize when such hidden branches occur.

After this, we consider explicit conditional statements, written directly by the user. These conditions are intrinsic to program logic, ensuring that certain computations occur only when specific criteria are met. Because they can't simply be eliminated, we revisit common approaches to expressing conditions and analyze their internal implementation. The aim is to compare their relative strengths and limitations, indicating which ones interact more favorably with SIMD execution.

Finally, we show that conditional statements can often be reformulated as algebraic operations. This transformation removes branching logic and restores SIMD's ability to operate on a uniform instruction stream. In fact, compilers may implement such transformations automatically when it's likely beneficial.

HIDDEN BRANCHING

Certain coding patterns result in **hidden branching**. These are conditional execution paths that the compiler inserts internally to handle the computation, rather than being explicitly written by the programmer.

TYPE INSTABILITY

Type instability is one source of hidden branching. When a function can return values of different types, Julia must generate separate execution paths to handle each possibility. These extra branches remain invisible in the source code, yet they still break the uniform instruction flow that SIMD depends on. The solution is simply to restore type stability, for which we can apply all the techniques explored in Chapter 8 of this book.

It's worth remarking, though, that type stability should always be a priority, regardless of any SIMD consideration: any attempt to achieve high performance is nearly impossible without it. Considering this, we'll proceed throughout the section assuming that type stability has been guaranteed.

BOUNDS CHECKING

Bounds checking in for-loops represents another source of hidden branching. This safety mechanism, enabled by default, verifies that indices stay within valid limits before executing each iteration.

The verification creates a subtle form of conditional execution, preventing the compiler from treating the loop as a uniform sequence of operations. Consequently, disabling these checks is **a prerequisite for the application of SIMD**. This is implemented by prepending the for-loop with `@inbounds`.

If bound checks remain in place, inserting `@simd` alone will rarely lead the compiler to emit SIMD instructions. In fact, merely adding `@inbounds` may be sufficient to enable SIMD on its own. In those cases, you'll notice that using `@inbounds @simd` rather than `@inbounds` won't yield any additional speedup. Both of these points are demonstrated below. [1]

NO MACROS

```
x = rand(1_000_000)

function foo(x)
    output = similar(x)

    for i in eachindex(x)
        output[i] = 2/x[i]
    end

    return output
end
```

```
julia> @btime foo($x)
  2.392 ms (3 allocations: 7.629 MiB)
```

@SIMD

```
x = rand(1_000_000)

function foo(x)
    output = similar(x)

    @simd for i in eachindex(x)
        output[i] = 2/x[i]
    end

    return output
end
```

```
julia> @btime foo($x)
  2.384 ms (3 allocations: 7.629 MiB)
```

@INBOUNDS

```
x = rand(1_000_000)

function foo(x)
    output = similar(x)

    @inbounds for i in eachindex(x)
        output[i] = 2/x[i]
    end

    return output
end
```

```
julia> @btime foo($x)
  1.226 ms (3 allocations: 7.629 MiB)
```

@INBOUNDS + @SIMD

```
x = rand(1_000_000)

function foo(x)
    output = similar(x)

    @inbounds @simd for i in eachindex(x)
        output[i] = 2/x[i]
    end

    return output
end
```

```
julia> @btime foo($x)
  5.382 ms (3 allocations: 7.629 MiB)
```

Overall, when signaling to the compiler that SIMD is appropriate, for-loops should be annotated with `@inbounds @simd` rather than `@simd` alone.

Broadcasting and For-Loops

Broadcasting in Julia automatically disables bounds checking and applies SIMD instructions. This often makes broadcast versions appear faster than a straightforward for-loop. However, **broadcasting is just compact notation for implementing for-loops**. As a result, a for-loop annotated with `@inbounds @simd` usually achieves a similar performance to its broadcast variant, as demonstrated below.

BROADCASTING

```
x      = rand(1_000_000)
foo(x) = 2 ./ x
```

```
julia> @btime foo($x)
  1.190 ms (3 allocations: 7.629 MiB)
```

FOR-LOOP (NO MACROS)

```
x      = rand(1_000_000)

function foo(x)
    output = similar(x)

    for i in eachindex(x)
        output[i] = 2/x[i]
    end

    return output
end
```

```
julia> @btime foo($x)
  2.384 ms (3 allocations: 7.629 MiB)
```

FOR-LOOP (OPTIMIZED WITH MACROS)

```
x      = rand(1_000_000)

function foo(x)
    output = similar(x)

    @inbounds @simd for i in eachindex(x)
        output[i] = 2/x[i]
    end

    return output
end
```

```
julia> @btime foo($x)
  1.191 ms (3 allocations: 7.629 MiB)
```

APPROACHES TO CONDITIONAL STATEMENTS

Certain workflows require conditional statements, making them an integral part of a program's logical flow. Because they can't be avoided, it's important to consider the most effective way to express them.

We've already examined several ways to introduce **conditional statements**. In the context of SIMD, a relevant distinction among these approaches is whether they evaluate conditional statements **eagerly or lazily**.

To illustrate, let's consider computing `1 + 1` only when certain condition `c` is met. A lazy strategy checks `c` first, performing the addition only if the condition is true. By contrast, an eager approach performs `1 + 1` immediately, regardless of whether `c` is satisfied. If the condition eventually evaluates to `false`, the result of `1 + 1` is simply discarded.

When a conditional is evaluated just once, a lazy approach is more performant because it avoids unnecessary work. In contrast, when that conditional appears inside a for-loop and is evaluated repeatedly, SIMD can apply the same operation across many elements simultaneously. In these cases, it's often advantageous to compute all branches upfront, selecting the relevant results only afterward. This eager strategy works particularly well when the branches involve inexpensive computations.

In Julia, whether a conditional expression is evaluated lazily or eagerly depends on how the code is written. Next, we'll examine this aspect in more detail.

IFELSE VS IF

The `ifelse` function in Julia follows an eager evaluation strategy, computing all possible outcomes before selecting the one that matches the condition. In contrast, the `if` construct favors lazy computations, where only the branch corresponding to the true condition is executed.

The following example demonstrates the computational difference between `if` and `ifelse`. It uses a reduction operation in which the elements to be summed are contingent on a condition. Because `ifelse` requires an argument for both the true and false cases, we return zero when the condition isn't met.

IF

```
x = rand(1_000_000)

function foo(x)
    output = 0.0

    for i in eachindex(x)
        if x[i] > 0.5
            output += x[i]/2
        end
    end

    return output
end
```

```
julia> @btime foo($x)
  1.312 ms (0 allocations: 0 bytes)
```

IF + SIMD

```
x = rand(1_000_000)

function foo(x)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        if x[i] > 0.5
            output += x[i]/2
        end
    end

    return output
end
```

```
julia> @btime foo($x)
  1.321 ms (0 allocations: 0 bytes)
```

IFELSE

```
x = rand(1_000_000)

function foo(x)
    output = 0.0

    for i in eachindex(x)
        output += ifelse(x[i] > 0.5, x[i]/2, 0)
    end

    return output
end
```

```
julia> @btime foo($x)
  1.221 ms (0 allocations: 0 bytes)
```

IFELSE + SIMD

```
x = rand(1_000_000)

function foo(x)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        output += ifelse(x[i] > 0.5, x[i]/2, 0)
    end

    return output
end
```

```
julia> @btime foo($x)
  349.996 µs (0 allocations: 0 bytes)
```

The example shows that `ifelse` doesn't automatically trigger SIMD execution, even though it uses eager evaluation. This is precisely why the code required the `@inbounds @simd` macros.

It's also worth noting that applying SIMD instructions with `ifelse` doesn't necessarily improve performance. The following example demonstrates this point.

IFELSE

```
x      = rand(5_000_000)
output = similar(x)

function foo!(output,x)
    for i in eachindex(x)
        output[i] = ifelse(x[i] > 0.5, x[i]/2, 0)
    end
end
```

```
julia> foo!($output,$x)
  56.122 ms (0 allocations: 0 bytes)
```

IFELSE + SIMD

```
x      = rand(5_000_000)
output = similar(x)

function foo!(output,x)
    @inbounds @simd for i in eachindex(x)
        output[i] = ifelse(x[i] > 0.5, x[i]/2, 0)
    end
end
```

```
julia> foo!($output,$x)
  50.983 ms (0 allocations: 0 bytes)
```

TERNARY OPERATOR

An alternative approach to expressing conditions is given by the ternary operation, whic follows the syntax `<condition> ? <action if true> : <action if false>`. Unlike the previous methods, this form doesn't enforce an eager or a lazy approach. Instead, it relies on compiler heuristics to determine which strategy will likely be faster for the specific application.

To illustrate this behavior, we revisit the same reduction example used earlier, now applying `@inbounds @simd` directly within each construct. In the following, the ternary operator chooses an eager strategy, thus evaluating both expressions before selecting the relevant branch.

IF

```
x = rand(1_000_000)

function foo(x)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        if x[i] > 0.5
            output += x[i]/2
        end
    end

    return output
end
```

```
julia> foo!($x)
  1.318 ms (0 allocations: 0 bytes)
```

IFELSE

```
x = rand(1_000_000)

function foo(x)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        output += ifelse(x[i]>0.5, x[i]/2, 0)
    end

    return output
end
```

```
julia> foo!($x)
  386.313 μs (0 allocations: 0 bytes)
```

TERNARY

```
x = rand(1_000_000)

function foo(x)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        output += x[i]>0.5 ? x[i]/2 : 0
    end

    return output
end
```

```
julia> foo!($x)
  392.661 μs (0 allocations: 0 bytes)
```

On the contrary, the ternary operator opts for a lazy approach in the following example.

IF

```
x = rand(1_000_000)

function foo(x)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        if x[i] > 0.99
            output += log(x[i])
        end
    end

    return output
end
```

```
julia> foo!($x)
  1.267 ms (0 allocations: 0 bytes)
```

IFELSE

```
x = rand(1_000_000)

function foo(x)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        output += ifelse(x[i] > 0.99, log(x[i]), 0)
    end

    return output
end
```

```
julia> foo!($x)
  11.496 ms (0 allocations: 0 bytes)
```

TERNARY

```
x = rand(1_000_000)

function foo(x)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        output += x[i]>0.99 ? log(x[i]) : 0
    end

    return output
end
```

```
julia> foo!($x)
  1.358 ms (0 allocations: 0 bytes)
```

It's worth noting that the method chosen by **the ternary operator isn't foolproof**, possibly choosing a less performant approach. This is illustrated below.

IF

```
x      = rand(5_000_000)
output = similar(x)

function foo!(output,x)
    @inbounds @simd for i in eachindex(x)
        if x[i] > 0.5
            output[i] = log(x[i])
        end
    end
end
```

```
julia> foo!($output,$x)
  79.400 ms (0 allocations: 0 bytes)
```

IFELSE

```
x      = rand(5_000_000)
output = similar(x)

function foo!(output,x)
    @inbounds @simd for i in eachindex(x)
        output[i] = ifelse(x[i] > 0.5, log(x[i]), 0)
    end
end
```

```
julia> foo!($output,$x)
  54.072 ms (0 allocations: 0 bytes)
```

TERNARY

```
x      = rand(5_000_000)
output = similar(x)

function foo!(output,x)
    @inbounds @simd for i in eachindex(x)
        output[i] = x[i]>0.5 ? log(x[i]) : 0
    end
end
```

```
julia> foo!($output,$x)
  80.596 ms (0 allocations: 0 bytes)
```

WHEN IS EACH APPROACH BETTER?

As a rule of thumb, conditional statements with **computationally-demanding operations are more likely to benefit from a lazy implementation**. In contrast, **branches comprising simple computations tend to benefit from an eager approach**. These are precisely the heuristics that typically drive the decision of the ternary operator.

To make this concrete, consider a case where only one branch performs a computation, and that computation is straightforward. Under these conditions, an eager evaluation combined with SIMD typically outperforms a lazy alternative. This is precisely the strategy adopted by the ternary operator.

IF

```
x              = rand(1_000_000)
condition(a)   = a > 0.5
computation(a) = a * 2

function foo(x)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        if condition(x[i])
            output += computation(x[i])
        end
    end

    return output
end
```

```
julia> foo!($x)
  1.296 ms (0 allocations: 0 bytes)
```

IFELSE

```
x            = rand(1_000_000)
condition(a)   = a > 0.5
computation(a) = a * 2

function foo(x)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        output += ifelse(condition(x[i]), computation(x[i]), 0)
    end

    return output
end
```

```
julia> foo!($x)
  412.153 μs (0 allocations: 0 bytes)
```

TERNARY

```
x            = rand(1_000_000)
condition(a)   = a > 0.5
computation(a) = a * 2

function foo(x)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        output += condition(x[i]) ? computation(x[i]) : 0
    end

    return output
end
```

```
julia> foo!($x)
  312.681 μs (0 allocations: 0 bytes)
```

The situation reverses when the conditional branch involves a computationally intensive operation. Here, a lazy strategy avoids performing expensive work unnecessarily, leading to better performance. Below, we illustrate a scenario like this, where the ternary operator chooses a lazy evaluation.

IF

```
x               = rand(2_000_000)
condition(a)    = a > 0.5
computation(a) = exp(a)/3 - log(a)/2

function foo(x)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        if condition(x[i])
            output += computation(x[i])
        end
    end

    return output
end
```

```
julia> foo!($x)
  39.390 ms (0 allocations: 0 bytes)
```

IFELSE

```
x               = rand(2_000_000)
condition(a)    = a > 0.5
computation(a) = exp(a)/3 - log(a)/2

function foo(x)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        output += ifelse(condition(x[i]), computation(x[i]), 0)
    end

    return output
end
```

```
julia> foo!($x)
  40.469 ms (0 allocations: 0 bytes)
```

TERNARY

```
x              = rand(2_000_000)
condition(a)   = a > 0.5
computation(a) = exp(a)/3 - log(a)/2

function foo(x)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        output += condition(x[i]) ? computation(x[i]) : 0
    end

    return output
end
```

```
julia> foo!($x)
  40.580 ms (0 allocations: 0 bytes)
```

VECTORS WITH CONDITIONS

In many situations, you may already have a vector of Boolean conditions defined. This can happen because the condition vector is part of the dataset itself, or because the same conditions will be reused throughout the code, making it worthwhile to compute and store them in advance.

Julia represents such vectors using either `Vector{Bool}` or `BitVector`. The latter is the default output for broadcast expressions like `x .> 0`. While `BitVector` is memory-efficient and can be faster in some contexts, it can also inhibit SIMD optimizations. When this occurs, converting the `BitVector` to a `Vector{Bool}` can improve performance. The example below illustrates this effect: even after accounting for the cost of converting the type, the version using `Vector{Bool}` achieves a shorter execution time.

BITVECTOR

```
x         = rand(1_000_000)
bitvector = x .> 0.5

function foo(x,bitvector)
    output     = similar(x)


    @inbounds @simd for i in eachindex(x)
        output[i] = ifelse(bitvector[i], x[i]/i, x[i]*i)
    end

    return output
end
```

```
julia> foo($x,$bitvector)
  13.156 ms (3 allocations: 7.629 MiB)
```

VECTOR{BOOL}

```
x         = rand(1_000_000)
bitvector = x .> 0.5

function foo(x,bitvector)
    output     = similar(x)
    boolvector = Vector(bitvector)

    @inbounds @simd for i in eachindex(x)
        output[i] = ifelse(boolvector[i], x[i]/i, x[i]*i)
    end

    return output
end
```

```
julia> foo($x,$bitvector)
  2.588 ms (5 allocations: 8.583 MiB)
```

No Vector Holding Conditions

When the vector of conditions isn't already available, the conclusions outlined above no longer apply. In those cases, the most efficient strategy to leverage SIMD is to avoid building a condition vector altogether. Instead, you should apply `ifelse` directly to scalar values inside the for-loop. This approach eliminates unnecessary memory allocations, while still allowing the compiler to generate SIMD instructions effectively. The following example demonstrates this point.

BITVECTOR

```
x = rand(1_000_000)

function foo(x)
    output     = similar(x)
    bitvector  = x .> 0.5


    @inbounds @simd for i in eachindex(x)
        output[i] = ifelse(bitvector[i], x[i]/i, x[i]*i)
    end

    return output
end
```

```
julia> foo($x)
  9.641 ms (7 allocations: 7.749 MiB)
```

VECTOR{BOOL}

```
x = rand(1_000_000)

function foo(x)
    output     = similar(x)
    boolvector = Vector{Bool}(undef,length(x))
        boolvector .= x .> 0.5

    @inbounds @simd for i in eachindex(x)
        output[i] = ifelse(boolvector[i], x[i]/i, x[i]*i)
    end

    return output
end
```

```
julia> foo($x)
  2.314 ms (5 allocations: 8.583 MiB)
```

NO VECTOR

```
x = rand(1_000_000)

function foo(x)
    output     = similar(x)



    @inbounds @simd for i in eachindex(x)
        output[i] = ifelse(x[i]>0.5, x[i]/i, x[i]*i)
    end

    return output
end
```

```
julia> foo($x)
  1.436 ms (3 allocations: 7.629 MiB)
```

ALGEBRAIC OPERATIONS AS COMPOUND CONDITIONS

Branches can often be eliminated by rewriting logical conditions in terms of simple algebraic identities. By doing so, we're eliminating the hurdle that impedes a direct application of SIMD.

To see how this works, let's first formalize Boolean logic in mathematical terms. Given a set $\{b_i\}_{i=1}^{n}$ where $b_i \in \{0, 1\}$:

- All conditions are satisfied when

$$\prod_{i=1}^{n} c_i = 1$$

- At least one condition is satisfied when

$$1 - \prod_{i=1}^{n} (1 - c_i) = 1$$

These identities translate directly into Boolean expressions. Specifically, given two Boolean scalars `c1` and `c2`:

- `c1 && c2` can be expressed as `Bool(c1 * c2)`,
- `c1 || c2` can be expressed as `Bool(1 - !c1 * !c2)`.

Once this strategy is adopted, control flow can be rewritten in a manner that's friendlier to vectorization. The following example illustrates this approach by replacing branching logic in a for-loop.

&&

```
x = rand(1_000_000)
y = rand(1_000_000)

function foo(x,y)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        if (x[i] > 0.3) && (y[i] < 0.6) && (x[i] > y[i])
            output += x[i]
        end
    end

    return output
end
```

```
julia> foo($x,$y)
  6.708 ms (0 allocations: 0 bytes)
```

&& (ALGEBRAIC)

```
x = rand(1_000_000)
y = rand(1_000_000)

function foo(x,y)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        if (x[i] > 0.3)  * (y[i] < 0.6)  * (x[i] > y[i])
            output += x[i]
        end
    end

    return output
end
```

```
julia> foo($x,$y)
  2.924 ms (0 allocations: 0 bytes)
```

||

```
x = rand(1_000_000)
y = rand(1_000_000)

function foo(x,y)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        if (x[i] > 0.3) || (y[i] < 0.6) || (x[i] > y[i])
            output += x[i]
        end
    end

    return output
end
```

```
julia> foo($x,$y)
  9.961 ms (0 allocations: 0 bytes)
```

|| (ALGEBRAIC)

```
x = rand(1_000_000)
y = rand(1_000_000)

function foo(x,y)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        if Bool(1 - !(x[i] > 0.3)  * !(y[i] < 0.6)  * !(x[i] > y[i]))
            output += x[i]
        end
    end

    return output
end
```

```
julia> foo($x,$y)
  2.806 ms (0 allocations: 0 bytes)
```

The following example does the same, but for the case of broadcasting.

&&

```
x        = rand(1_000_000)
y        = rand(1_000_000)

foo(x,y) = @. ifelse((x>0.3) && (y<0.6) && (x>y), x, y)
```

```
julia> foo($x,$y)
  15.562 ms (3 allocations: 7.629 MiB)
```

ALGEBRAIC &&

```
x        = rand(1_000_000)
y        = rand(1_000_000)

foo(x,y) = @. ifelse((x>0.3) *  (y<0.6) *  (x>y), x, y)
```

```
julia> foo($x,$y)
  1.748 ms (3 allocations: 7.629 MiB)
```

||

```
x        = rand(1_000_000)
y        = rand(1_000_000)

foo(x,y) = @. ifelse((x>0.3) || (y<0.6) || (x>y), x, y)
```

```
julia> foo($x,$y)
  11.352 ms (3 allocations: 7.629 MiB)
```

ALGEBRAIC ||

```
x        = rand(1_000_000)
y        = rand(1_000_000)

foo(x,y) = @. ifelse(Bool(1 - !(x>0.3) * !(y<0.6) * !(x>y)), x, y)
```

```
julia> foo($x,$y)
  1.459 ms (3 allocations: 7.629 MiB)
```

Notice that such algebraic transformations may be performed automatically by the compiler. Whether this occurs depends on how the code is written. For example, revisiting the case of for-loops, the compiler performs the transformation automatically when expressing the condition through a separate function.

&&

```
x              = rand(1_000_000)
y              = rand(1_000_000)


function foo(x,y)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        if (x[i] > 0.3) && (y[i] < 0.6) && (x[i] > y[i])
            output += x[i]
        end
    end

    return output
end
```

```
julia> foo($x,$y)
  6.732 ms (0 allocations: 0 bytes)
```

&& (ALGEBRAIC)

```
x              = rand(1_000_000)
y              = rand(1_000_000)


function foo(x,y)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        if (x[i] > 0.3)  * (y[i] < 0.6)  * (x[i] > y[i])
            output += x[i]
        end
    end

    return output
end
```

```
julia> foo($x,$y)
  2.727 ms (0 allocations: 0 bytes)
```

AUTOMATIC TRANSFORMATION

```
x              = rand(1_000_000)
y              = rand(1_000_000)
condition(a,b) = (a > 0.3)  * (b < 0.6)  * (a > b)

function foo(x,y)
    output = 0.0

    @inbounds @simd for i in eachindex(x)
        if condition(x[i],y[i])
            output += x[i]
        end
    end

    return output
end
```

```
julia> foo($x,$y)
  2.791 ms (0 allocations: 0 bytes)
```

FOOTNOTES

[1] Recall that the compiler may automatically disable bounds checking in simple scenarios. For instance, this is the case when only `x` is indexed and `eachindex(x)` is employed as the iteration range. Instead, explicit use of `@inbounds` becomes necessary when both `x` and another vector indexed, as in the examples we introduce below.